OPTO

Operational manual for Topspin

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1. Introduction:

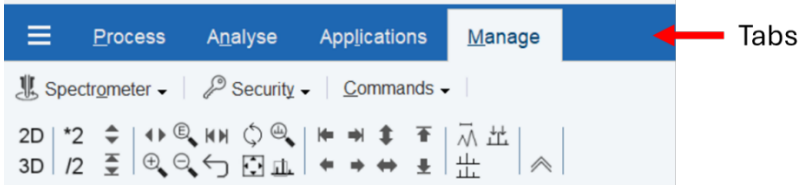
This manual is included with the installer and can be found in the unzipped folder under the file name "`opto-help.pdf`". After successful installation, this manual can be accessed by pressing the 'Help' button located under the OPTO tabs within TOPSPIN. The manual provides a comprehensive guide to all OPTO settings and capabilities, including an example run to demonstrate how OPTO can be utilized for TOPSPIN.

For the latest version of OPTO for Topspin, go to https://git.doit.wisc.edu/nmrfam-public/opto-releases/-/blob/main/README-TOPSPIN.md

2. Installation Guide:

a) Download the most recent `opto-topspin.zip` file from Barry for now.
b) Launch a terminal.
c) Use the `cd <directory-path>` command to navigate to the directory that contains `opto-topspin.zip`.
d) Execute the `unzip opto-topspin.zip` command to extract the `opto-topspin.zip` file.
e) Switch to the directory that was just extracted.
f) Install opto by running the `sudo ./install.sh <topspin-version>` command.
    a. For instance: `sudo ./install.sh topspin3.6.2`
g) If you wish, you can remove the files that were unzipped.

3. Adding Shortcuts for OPTO:

a) Launch the TOPSPIN software.
b) Right-click on the empty space on the Tabs section as shown below.
    a. 

c) Choose '`Add new Tab`'.
d) Input the following:

    Tab ID: `OPTO`, Tab Text: `Opto`, Flowbar File: `opto.prop`

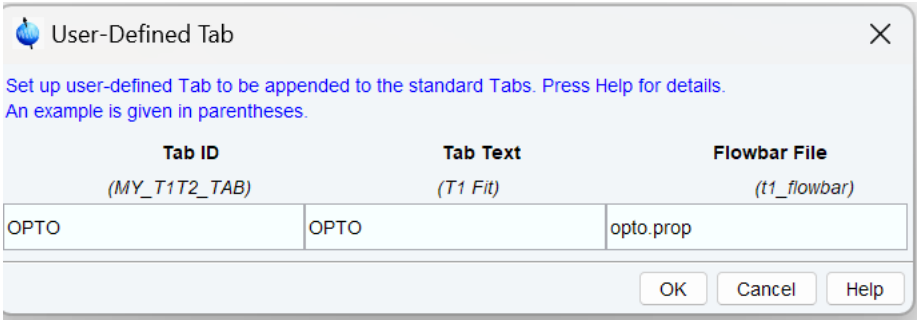


Figure: Screen shot showing the addition of shortcut to topspin toolbar.

4. Initial Setup:

Shimming: OPTO utilizes a text file named `opto-shim.dat` to define the shim names, their limits, and step size. A default file is included with the installation and can be found at `$topspin-home/exp/stan/nmr/lists/vd`. This file can be modified by running the `edlist opto-shim.dat` command in the topspin command line.

Parameter Optimization: OPTO uses a parameter file located at `$topspin-home/opto` folder. Default parameter file are provided with the installation for hC CP pulse program as an example. Initially, users should copy this file with a new name to the `$topspin-home/opto` folder and then modify this file to meet their needs by following these steps:

a) Open the experiment that needs to be optimized.
b) List the parameters in the text files along with their minimum, maximum, and step size.
c) Include all the relevant parameters that are defined in your pulse program, their global limits, and a reasonable step size.

The `$topspin-home/opto` folder should contain parameter files tailored to each of your pulse sequences of interest.

4.1 Selecting the region of interest for the integration:

OPTO utilizes an integration file `intrng` to calculate the total area of a specified region. To save a region of interest:

a) Acquire a 1D spectrum.
b) Access the integration module by entering `.int`
c) Clear any existing integral regions.
d) Define the region of interest.
e) Export the region to an `intrng` file by clicking the export button displayed below.

This process ensures the region of interest is saved to the `intrng` file. Note that OPTO reads full spectral width by default if no `intrng` file exists. If multiple regions are stored in the

intrng file, OPTO only processes the first region. It will show the following message in the topspin terminal if these is not intrng file exists.

```
No intrng file found!!!!! using the full spectra width for
integration!!!!!! save the intrng file to select the region for
intergration
```



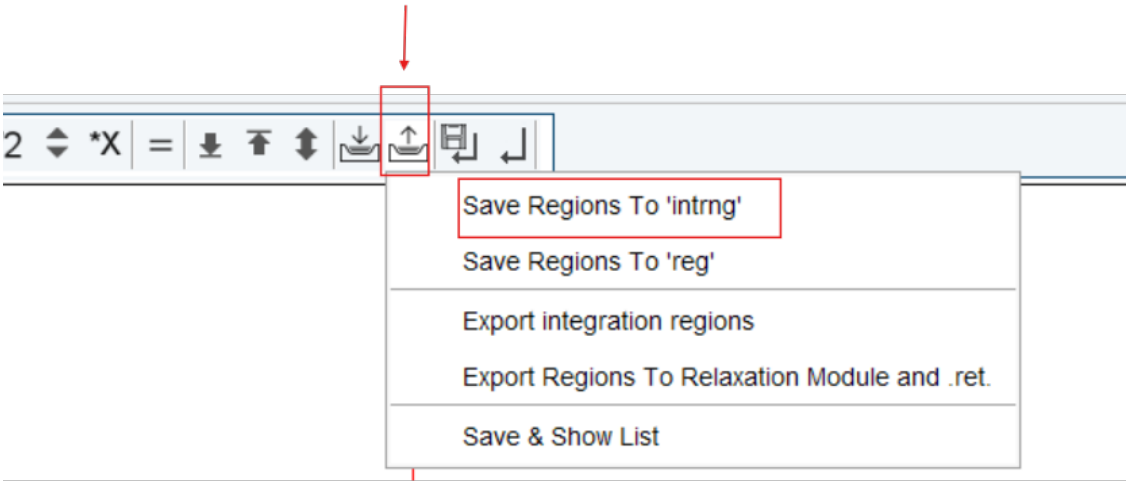


Figure 2: Screen shot showing the method to save intrng file with the selected region of interest for the integration.

4.2 Shape Optimization for Cross Polarization with OPTO:

OPTO can optimize the shape parameters for a given Ramp on selected nuclei. Ramp shapes are defined by two key parameters: BETA and DELTA.

4.2.1 Definition of Variables and Conventions:

- **BETA:** Influences the overall amplitude and the steepness of the modulation curve.
- **DELTA:** Affects the angular coefficient in the tangent function, influencing the curvature of the modulation profile.

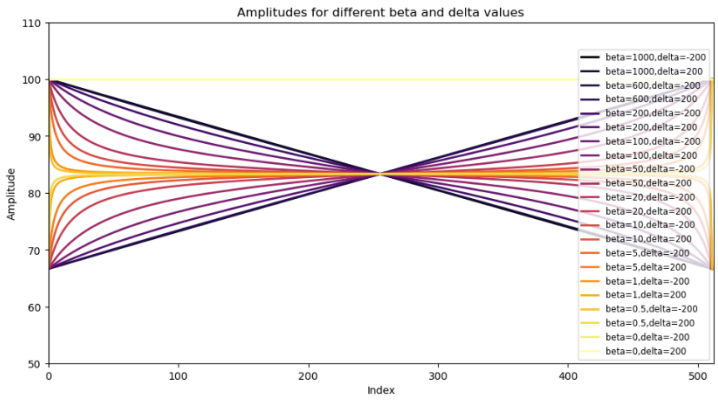
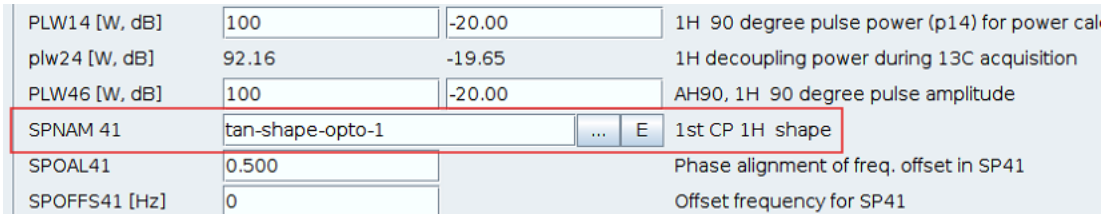
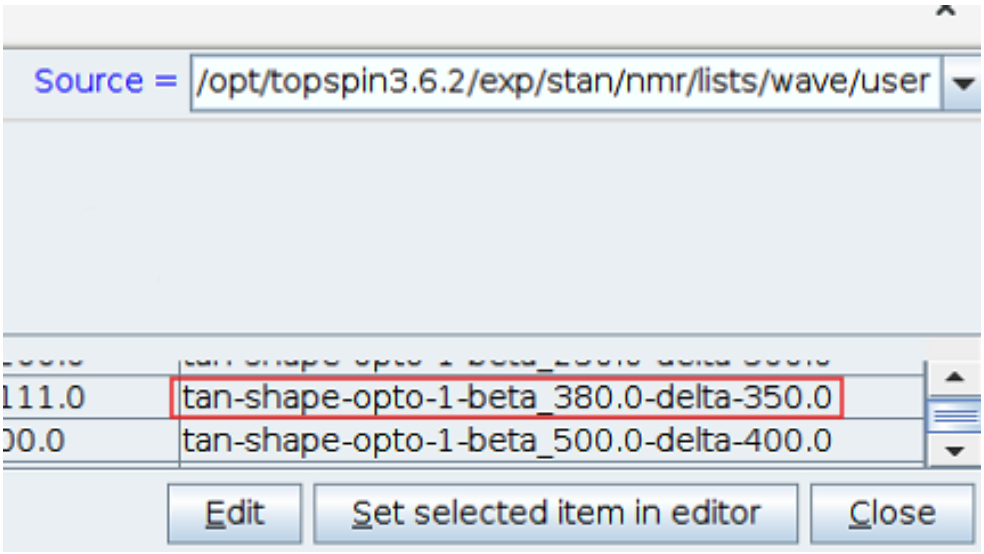


Figure: Tangential ramp shapes with different BETA and DELTA values

- **Index Number:** Each pair, BETA and DELTA, is assigned an index number (e.g., BETA 1 and DELTA 1, BETA 2 and DELTA 2). This facilitates the modulation of different shapes within the same pulse program in a single optimization run.
- **Naming Convention:** Each tangent shape requires both BETA and DELTA values, along with their index numbers, to be included in the parameter file. For example, BETA 1 and DELTA 1 would generate a shape saved as **tan-shape-opto-1** in the corresponding user folder at **$topspin-home/exp/stan/nmr/lists/wave/user**. The number in the shape file name represents the index number. By altering the index numbers in the BETA and DELTA names, any number of shapes can be generated. Each shape file must be selected in AcqPars to optimize the shape for a given nucleus.



- **Saving the Shape File:** Upon successful completion of the optimization run, OPTO saves the final optimized shape file named **tan-shape-opto-x-beta_y-delta_z**, where x is the index number, and y and z represent the corresponding BETA and DELTA values, respectively. After the run, users need to load this shape to conduct further experiments.



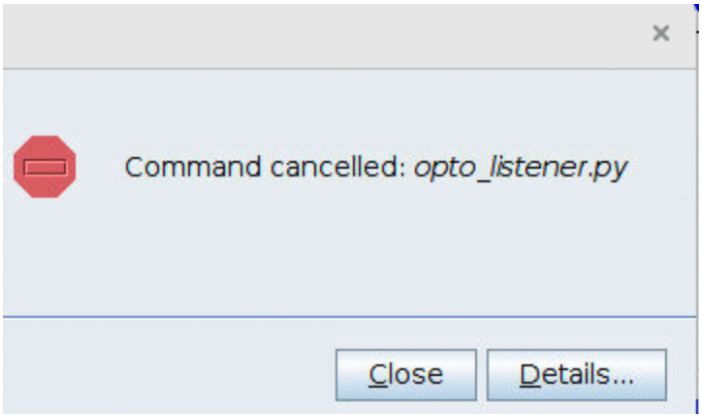
5. Opening OPTO in TOPSPIN:

- To open OPTO, click on the `'open-opto'` button.
- A popup window will appear where you need to select the parameter file.
- Once the parameter file is selected, the OPTO window will appear.

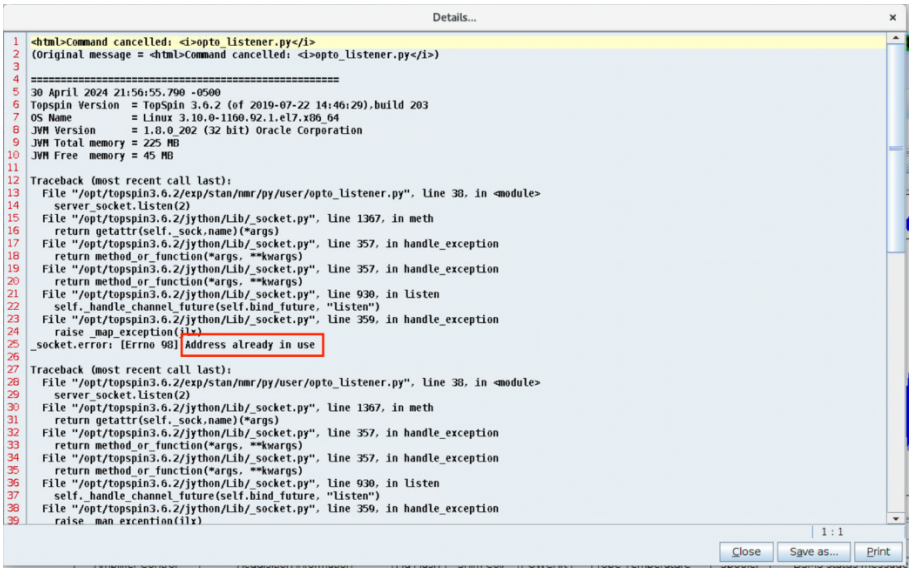
Note: If an error message appears in the TOPSPIN window (see figure below), TOPSPIN must be restarted before starting OPTO. This is necessary if the previous session of OPTO was not closed properly (see below).

5.1 Troubleshooting OPTO start-up issues:

If the following message appears while starting OPTO, click on the 'Details' button to see more information.



If this message appears, it may indicates that OPTO was not properly closed from the previous session. Click on `Details.`



After clicking on details if you see the above message, In this case TOPSPIN software needs to be restarted.

6. Closing OPTO in TOPSPIN:

- To close OPTO, first close the OPTO window.
- Then click on the 'close-opto' button in TOPSPIN.
- OPTO can also be opened by typing the command **`open-opto`** in the TOPSPIN command line and closed with the command **`close-opto`**.
- When executing the **`close-opto`** command, the TOPSPIN terminal should display the message "`Closing server socket…..`" If this message does not appear in the terminal window, TOPSPIN needs to be restarted before starting another OPTO session.

6.1 Additional Information:

- OPTO is only usable with one-dimensional spectra. It can be started when an active 1D dataset is open, and this dataset will remain active for the entire session until the **`close-opto`** command is executed or the 'close-opto' button is pressed.
- This action will close the communication socket, allowing a new session of OPTO to be opened on the same or a different dataset as needed.

- If the same session continues, OPTO will always use the initial dataset from which it was opened, and all displayed FID/Linewidth/intarea plots are saved per session. For a given session, OPTO will plot the relative intensity/FID plots for all the runs executed within that session.
- If the session is closed using **`close-opto`**, the session will be cleared.

7. Accessing OPTO Settings:

- OPTO settings can be accessed by clicking on the 'OPTO-settings' button or by executing the **`opto-settings`** command in the TOPSPIN command window.
- This action opens a window to choose settings as described below.

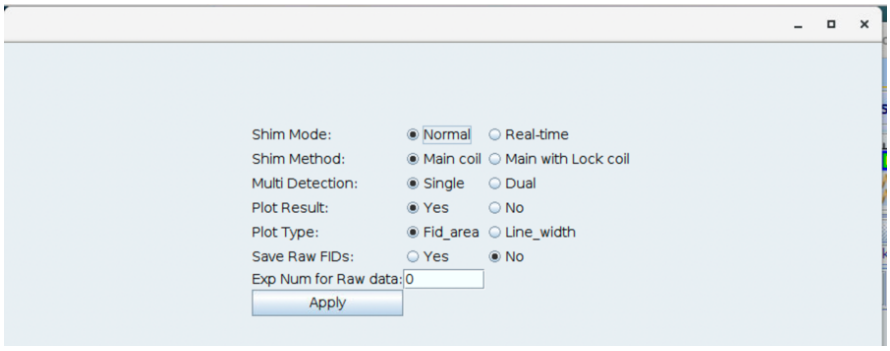


Figure: OPTO settings window showing various functionalities and their selection

8. OPTO Capabilities and Settings:

8.1 Shim Mode:

1. **Normal:**
   - In this mode, OPTO runs the active experiment with a specified number of scans and proceeds only after the experiment is complete before starting the next experiment. This mode is recommended for samples with low signal-to-noise ratio (SNR) for shimming.
2. **Real-time:**
   - In this mode, OPTO reads the Free Induction Decay (FID) after each scan without reloading all parameters, similar to the 'gs' mode in TOPSPIN. High SNR samples are required for this mode as it uses just one scan to read the FID.

8.2 Instructions to Set Up Shimming in Real-Time Mode:

a) Load the pulse sequence **`opto-realtime-hpdec`** (provided and placed in **`$topspin-dir/exp/stan/nmr/pp/user`**).
b) Adjust parameters as necessary.
c) Set the number of scans to 32.
d) Start the experiment with the **`zg`** command.
e) Use the **`tr`** command to transfer the data.
f) After a few scans, use the **`tr`** command again.
g) Observe that the spectral intensity should remain consistent.
h) Open the `'OPTO-settings'` window.
i) Select `'Shim Mode: Real-time'`.
j) Click the `'Apply'` button and confirm in the next window.

Note: Real-time shim mode will begin the acquisition with a very large number of scans (e.g., 100,000) and will display a lengthy experiment time. After the OPTO run finishes, it will stop automatically.

8.3 Shim Methods:

1. **Main Coil:**
   - Designed to shim using one coil (typical configuration).
2. **Main Coil with Lock Coil:**
   - This method is intended to shim two coils simultaneously if the probe features a main coil and a separate lock coil.

Instructions for Setting Dual Coil Shim:

a) Run an experiment to collect the signal from the main coil (e.g., 13C signal from adamantane).
b) Run another experiment to collect the 2H spectra.
c) In the `'OPTO-settings'` window, select `'Main Coil with Lock'`.
d) Enter the experiment number for the lock coil.
e) Enter a scaling factor if different from the default.
f) Click `'Apply'`.

8.4 Multi-Detection (Under Development):

- This setting allows for either single or dual detection; however, it is currently inactive and the manual is still under construction.

8.5 Plotting Results:

- OPTO plots results in the TOPSPIN window. Users can choose between plotting FID area or linewidth for shimming purposes.
- For parameter optimization, OPTO will plot the integrated intensity only.

8.6 Saving Raw FIDs:

- If enabled, you will be prompted to enter the experiment number to save the raw FID.

Instructions for Saving Raw FIDs:

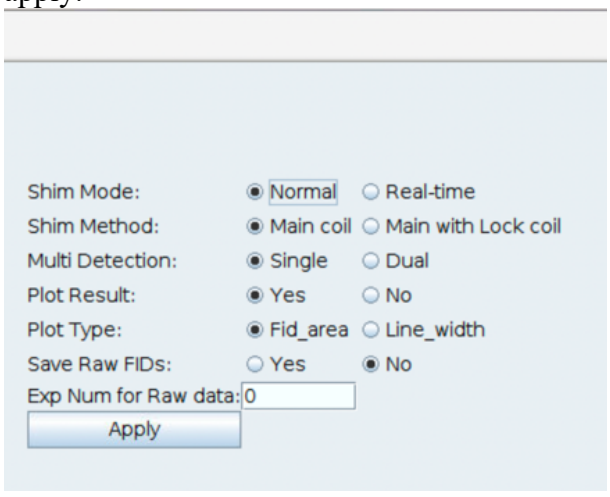
a) Copy the current active experiment to a new another experiment by typing the **`edc`** command in the TOPSPIN terminal.
b) Convert this 1D dataset into a 2D dataset by clicking on the 2D icon in the AquaPar parameter window(`eda`).
c) With the 2D dataset active, type **`1S TD`** and enter 1024 or a larger number in the indirect dimension based on the number of FIDs you want to save. Here the status parameter needs to be changed to a larger number to save more FIDs into the file.
d) Enter the newly created number into the raw Fid save exp number window

Note: If real-time mode shimming is selected, no raw data will be saved.

9. Running OPTO for Shimming:

Prerequisites: Have a valid `opto-shim` and a `parameter file.dat` file, standard sample loaded and MA adjusted, initial spectra is acquired.

a. Open the experiment and acquire a initial spectrum.
b. Click on `opto-settings` button or type `opto-settings` in the topspin command line
c. Select the settings as required, leave them as it is for default settings and click apply.
d.



e. Click on the `open-opto` button or type `open-opto` .
f. Select the parameter file from the pop-up menu.

g. At this point, the OPTO-window will appear with all the current shims loaded into the OPTO window. NOTE: it may take several seconds to show the OPTO window.
h. In pass 1 select the first order shims (`x1, y1, z1`). Click on add pass and enter number 2, this will create pass 2
i. In pass 2 select the second order shims (`xy, xz, yz, z2, x2y2`).). Click on add pass and enter number 3, this will create pass 3

j. In pass 3 select the third order shims (`x3, y3, z3, xz2, yz2, zxy, zx2y2`).
k. Change the number of iterations to 20 , range to 20, and tolerance 0.01 for experiment.
l. Click Run job.
m. This will start running the job, and the results will be plotted with in the topspin window and can also be seen on the topspin terminal.
n. Additionally user can choose between FID_area vs linewidth to plot in topspin to plot the results.
o. For additional improvements to the shimming, run the above passes 1-3 twice more after they have converged with setting the acquisition time to 200 ms for the first replicate, and then set the acquisition time to 400 ms for the last replicate.

10. Running OPTO for Magic Angle Setting:

a) **Prepare the Sample:**
   a. Insert a standard sample suitable for setting the magic angle, such as KBr. Spin the sample to anticipated speed. Load previously acquired 1D spectrum and acquire a initial spectra. Create new experiment (`edc`). This methos is useful if you have high SNR sample. In case of the low SNR samples it is recommend to use the normal method.
b) **Load the Pulse Program:**
   a. Load the **`opto-realtime-hpdec`** pulse program.
c) **Set Nucleus and Parameters:**
   a. Change the observed nucleus to 79Br.
   b. Enter the necessary power levels and pulse widths.
   c. Set the decoupling power (PLW12) to 0.
d) **Adjust OPTO Settings:**
   a. Open the opto settings by clicking on **`settings`** button in the Opto tab and select `'Real-time'` in the `shim mode` section.
   b. Choose the `FID_area` to plot.
   c. Click on `'Apply'`.
e) **Start OPTO:**
   a. Open OPTO by clicking on the `'open-opto'` button.
   b. Start the acquisition when prompted by clicking 'OK'.

c. Select the parameter set **`MAS-opto-parameter.dat`** . This file is included in with the installation and used a dummy parameter to run the experiment. Here we using D59 as dummy parameter, can be changed to any other unused parameter if needed.

f) **Optimize Parameters**
   a. Once the OPTO window is open, navigate to the parameter tab.
   b. Select '`Grid`' and '`D59`' to optimize.
   c. Click on '`Run job`'.
   d. Change the number of steps to `5000` and click '`OK`'.

g) **Monitor and Adjust:**
   a. OPTO will start and display a plot of the FID area in the TOPSPIN window, updating the intensity with every scan.
   b. Adjust the magic angle according to the probe hardware instructions to maximize the intensity of the FID area.

h) **Complete the Process:**
   a. Once maximum intensity is reached, click on '`Abort nicely`' to stop the process. This will end the OPTO job and stops the experiment.
   b. Click OK on the opto complete message
   c. Close the OPTO window and click on the '`close-opto`' button in `Opto` tab with in TOPSPIN.

i) **Verify the Setting:**
   a. Acquire a 1D 79Br spectrum with 64 scans to further check the magic angle setting using the default experiment used in the initial setup (step a).

11. Running OPTO for CP:

a) **Sample Preparation:**
   a. Load the sample (in this example, NAV spinning at 13.33 kHz).

b) **Experiment Setup:**
   a. Open the experiment with default parameters (for example, "`cp`" from the Bruker library).

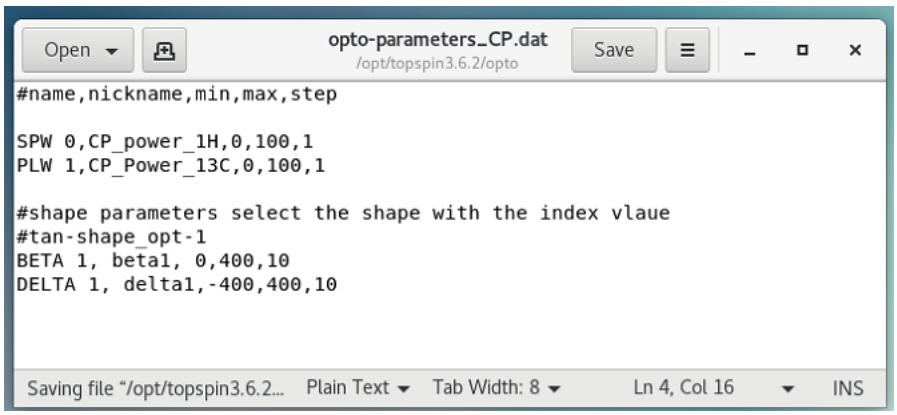
c) **Initial Spectrum Acquisition:**
   a. Acquire an initial spectrum with appropriate decoupling, acquisition time, and number of scans. Here the user is expected to have a theoretical starting point get an initial CP signal. A previously optimized CP experiment can be a good starting point here.

d) **Select Region of Interest:**
   a. Select the region of interest and save the region to an **`intrng`** file.

e) **Edit Parameter File:**
   a. Open the **`opto-parameters.dat`** file located in **`$topspin-dir/opto/.`**
   b. Edit the file to enter the parameters you want to optimize. Save it under a different filename in the same folder (example: "`opto-parameters_CP.dat`"). Unused parameters can be removed or commented out.

Note: Ensure that all parameter names (`P, D, CNST, SPW, L, PLW, PLdB`) have a space before the corresponding number.

f) **Set BETA and DELTA:**
   a. Include `BETA 1` and `DELTA 1` and accept the default values provided in the file.
   b. Save the file.
   i.



g) **Select CP Ramp Shape File:**
   a. Go to `AcquPars` and select the CP ramp shape file (`spnam0`) for 1H to be "`tan-shape-opto-1`"
   b. **Note:** If this is the first run using OPTO, create a shape file named as "`tan-shape-opto-1`."
   i.



h) **Open OPTO:**
   a. Open OPTO by clicking on the `'open-opto'` button or by typing **`open-opto`** in the TOPSPIN command line.

i) **Load Parameters File:**
   a. Select the parameters file (in this example, "`opto-parameters_CP.dat`") and click OK.
   b. Navigate to the `parameters` tab

j) **Adjust Parameters:**
   a. The OPTO window will show up with the parameters and their limits as specified in the parameters file.
   b. Select the ^{13}C and ^{1}H power for CP (`SPW 0, PLW 1`) in the first pass by checking on the boxes next to these parameters and adjust the limits as needed.
   c. Add a pass by clicking on the `add pass`
   d. In the second pass, select `BETA 1` and `DELTA 1` along with the ^{13}C and ^{1}H power levels.
   e. In the third pass select all four parameters `SPW 0, PLW 1, BETA 1` and `DELTA`
   f.

k) **Run Optimization:**
   a. Use the default settings for the tolerance and range change them if needed.

b. Click on '`Run job`'.
c. The results will be plotted in the TOPSPIN window as the run progresses.

l) **Completion and Application:**

a. After completion, a completion message will appear along with the optimized shape file. Click OK and close
b. Go to `AcquPars` and select the final shape file to proceed with further experiments.